Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Parameter updates can reinforce or cancel one another even when training loss continues to decrease. An exact finite-time decomposition of stochastic gradient descent (SGD) displacement separates gradient magnitudes from interactions across all temporal lags. Consecutive-gradient cosine provides a local directional diagnostic, while normalized displacement measures cumulative retention of traveled distance. Local expansions connect these quantities to curvature and sampling noise without identifying alignment sign with stability or objective progress. Full-batch experiments exhibit gradient reversal alongside continued loss reduction. Matched mini-batch continuations show that sampling changes both gradient estimates and the paths traversed through the objective. In Adam, raw-gradient dynamics provide information about impending loss excursions beyond recent loss history, even though the optimizer transforms those gradients before updating the parameters. Controlled full-batch experiments support this claim through exploratory forward-time prediction, while frequent threshold false alarms limit a standalone warning rule. The resulting framework supports the formulation and evaluation of informed learning-rate interventions; it does not prescribe an alignment-maximizing schedule.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Learning-Rate Scheduling**

I. Introduction

Neural scaling laws have ushered in new paradigms for interpreting deep learning dynamics. By establishing that performance gains follow predictable behavior relative to model size, dataset scale, and compute, stochastic optimization methods have been pushed toward increasingly extended training horizons while operating under tight efficiency constraints [1]–[4]. In this regime, optimization stability becomes an important practical requirement, since unstable training may result in slow convergence and high sensitivity to hyperparameter choices. In practice, stability is often maintained through a set of heuristics, such as momentum, batch-size scaling, normalization, and learning-rate schedules [5]–[9]. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

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Classical local analyses relate the permissible step size to curvature in sharp directions of the loss [10], [11]. Stochastic extensions, including the break-even point framework, also account for curvature variation induced by sampling [12]. The Edge-of-Stability literature shows that deep networks can continue to make progress while oscillating near, or beyond, the nominal sharpness threshold [13]–[15]. These results motivate a complementary diagnostic question: how do the magnitudes and directions of successive gradients combine into parameter motion along the observed training trajectory? Gradient information already computed during optimization makes this question accessible without estimating the full Hessian spectrum or gradient-noise covariance at every update.

To examine these interactions, we describe parameter motion through gradient magnitudes and directional alignment. Starting from an untrained reference, training changes the parameters through which the model expresses what it has learned. Displacement provides a geometric description of that change; it does not quantify retained knowledge or establish whether the change improves the objective. Positive and negative gradient interactions respectively reinforce and cancel components of the accumulated motion. Their effect must be evaluated at the relevant temporal horizon and distinguished from loss reduction.

We develop an exact finite-time decomposition of SGD displacement that retains all cross-step interactions. Consecutive-gradient cosine supplies a local online diagnostic, while normalized displacement measures the fraction of accumulated step length retained as net motion. Local expansions relate gradient changes to directional curvature and sampling noise. The distinction between these quantities is consequential: a trajectory can reverse successive gradient directions while continuing to reduce loss, and sampling can change both the measured alignment and the subsequent path.

The same distinction becomes essential under adaptive optimization. Adam already computes raw gradients and moment estimates [16]; its parameter updates combine temporal averaging, coordinate scaling and the learning rate. Monitoring these successive quantities can reveal changes that a single alignment statistic would obscure. Raw-gradient dynamics provide information about impending loss excursions beyond recent loss history, even though Adam transforms those gradients before updating the parameters. We examine this claim through controlled training and forward-time comparisons, distinguishing predictive information from a demonstrated improvement in learning-rate decisions.

Our contributions are:

- A displacement-based account of local and cumulative

gradient interaction, with explicit connections to curvature, sampling noise and finite-step loss changes;
- Controlled full-batch and mini-batch experiments that distinguish directional reversal, displacement retention and objective progress;
- A practical extension to Adam that separates raw-gradient information, momentum and adaptive directions, supported by controlled training and an exploratory assessment of loss-spike warnings beyond recent loss.

The remainder of the paper is organized as follows. Section II reviews training-stability perspectives. Section III develops the displacement formulation and its diagnostics. Section IV examines full-batch GD and mini-batch sampling. Section V extends the interpretation to Adam and outlines the empirical requirements for informed scheduling. Section VI concludes.

II. Background: Stability Analysis of SGD

Training stability concerns how perturbations and loss evolve under repeated updates. Overshoot and oscillation are useful symptoms, but can also accompany sustained objective progress. Understanding these dynamics requires examining both the *local curvature* of the loss landscape and the *stochasticity* introduced by mini-batch sampling [10], [11], [13].

A. Local Curvature of the Loss Function and Stability

Training can be interpreted as a discrete-time dynamical system in which the local curvature of the loss function determines the stability of the updates [10], [11]. The curvature indicates how sensitive the loss is to parameter changes in different directions: when curvature is small, updates tend to produce smooth descent toward a minimum, whereas in directions of high curvature the same step can overshoot the minimum and induce oscillatory behavior [10], [11], [17]. For this reason, classical analyses of gradient descent relate stability to the spectrum of the Hessian of the loss function, whose eigenvalues quantify directional curvature and indicate how sensitive optimization is to the choice of learning rate [10], [11].

Under this view, let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, the SGD update is

$$\theta_t = \theta_{t-1} - \eta\, g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$, where $\mathcal{L}$ denotes the full training objective. Defining $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of the deterministic update map is

$$J_t = I - \eta\, H_t. \tag{2}$$

Consider a small perturbation e_{t-1} around θ_{t-1}. Under the local linearization of the full-batch GD update, this perturbation evolves as

$$e_t \approx J_t e_{t-1} = (I - \eta H_t) e_{t-1},$$

In a locally quadratic regime near a minimum, the Hessian is approximately constant and admits eigenpairs (λ_i, v_i). Decomposing the perturbation as $e_t = \sum_i \delta_t^{(i)} v_i$, each eigendirection evolves independently according to

$$\delta_t^{(i)} = (1 - \eta\lambda_i)\, \delta_{t-1}^{(i)}.$$

Thus, the factor $1 - \eta\lambda_i$ determines whether perturbations along direction v_i decay or grow. For a positive-definite Hessian, a classical necessary and sufficient condition for linear stability is

$$\max_i |1 - \eta\lambda_i| < 1 \;\Leftrightarrow\; \eta\,\lambda_{\max}(H_t) < 2, \tag{3}$$

which means that all perturbation components contract over time [10], [11]. In other words, if $0 < \eta\lambda_i < 1$, the perturbation decays monotonically along v_i; if $1 < \eta\lambda_i < 2$, it alternates sign while shrinking, producing oscillations around the minimum; and if $\eta\lambda_i > 2$, it alternates sign while growing, leading to divergence. In over-parameterized networks, however, the Hessian spectrum is typically highly anisotropic, with a small number of outlier directions carrying most of the sharpness, while the bulk remains near zero [18], [19]. Although this suggests monitoring $\lambda_{\max}(H_t)$ as a stability diagnostic, estimating leading Hessian eigenvalues online in large neural networks is computationally expensive, and mini-batch estimates provide only noisy, pointwise proxies of the local geometry.

B. Mini-Batch Stochasticity

A complementary line of work studies SGD through the lens of stochastic optimization, interpreting the mini-batch gradient as a noisy approximation of the full gradient computed over the entire dataset [20]–[23]. In this sense, each SGD step combines a deterministic descent component with a zero-mean random fluctuation induced by sampling, which covariance provides information about how noise is distributed across parameter-space directions and how it may affect trajectory stability. Formally, under fresh, conditionally unbiased sampling, we write the mini-batch gradient as

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &\approx \frac{1}{b}\,\Sigma(\theta_{t-1}), \end{aligned} \tag{4}$$

where ξ_t is the sampling noise, b is the batch size, and $\Sigma(\theta_{t-1})$ denotes the covariance matrix of per-sample gradients around the full gradient. At the current parameters, write $h_i := \nabla\ell_i(\theta_{t-1})$, where ℓ_i is the loss for sample i, and $\mu := \frac{1}{N}\sum_{i=1}^N h_i = \nabla\mathcal{L}(\theta_{t-1})$, where N is the dataset size. The covariance matrix is

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}\big(h_i - \mu\big)\big(h_i - \mu\big)^\top, \tag{5}$$

and the scalar dispersion of per-sample gradients around the full gradient is given by the trace of $\Sigma(\theta_{t-1})$

$$\mathrm{tr}(\Sigma(\theta_{t-1})) = \frac{1}{N}\sum_{i=1}^{N}\|h_i - \mu\|^2.$$

This makes clear why $\Sigma(\theta_{t-1})$ is generally not a viable online diagnostic: computing it requires access to the full gradient μ and all per-sample gradients $\{h_i\}_{i=1}^N$ at each training step, which is impractical and incurs at least a full-pass cost per step.

C. Curvature–Noise Interaction

The *break-even point* framework of Jastrzebski et al. [12] is an attempt to extend the classical curvature-based stability condition to SGD. The central idea is to analyze stability along the sharpest direction of the loss landscape while explicitly accounting for mini-batch stochasticity. More precisely, the framework proposes a one-dimensional necessary-and-sufficient condition for SGD stability at iteration t,

$$(1 - \eta\lambda_{\max}(H_t))^2 + s_t^2 \frac{\eta^2(N-b)}{b(N-1)} \;\leq\; 1, \tag{6}$$

where $\lambda_{\max}(H_t)$ is the top Hessian eigenvalue, N is the dataset size, b is the batch size, and s_t^2 denotes the variance across training examples of the curvature measured along the leading Hessian direction. Large s_t^2 means different mini-batches see substantially different local curvature, while small s_t^2 means they see similar curvature. In this sense, Eq. (6) can be read as a stochastic extension of the deterministic stability condition in Eq. (3), so that the first term captures curvature-induced amplification, while the second captures the destabilizing effect of mini-batch stochasticity.

Under this definition, the break-even point is defined as the first iteration at which the left-hand side of Eq. (6) reaches 1. Intuitively, it marks the end of an early phase during which the choice of learning rate and batch size can strongly shape the later optimization trajectory. In this sense, a central conjecture of this framework is that, prior to this point, hyperparameters largely determine the maximum curvature and stochasticity levels encountered later in training, thereby setting the regime in which the subsequent dynamics unfold.

Although the local analysis simplifies both the interpretation and the mathematical treatment of the stability problem, it is subject to well-known limitations. For instance, the Edge-of-Stability (EoS) phenomenon challenges the local-quadratic picture even in deterministic gradient descent (GD). Across architectures and tasks, full-batch GD often exhibits progressive sharpening, with the largest eigenvalue of the Hessian, $\lambda_{\max}(H_t)$, rising until it hovers near (and sometimes above) the nominal threshold $2/\eta$ (Eq. 3), while optimization continues to make progress despite short-term non-monotonicity [13]–[15]. Thus, successful training can coexist with local linear instability for extended periods, indicating that sharpness alone does not fully explain and determine trajectory-level behavior.

Together, these results clarify both the value and the limitation of curvature–noise interaction criteria. Quantities such as the leading Hessian eigenvalue $\lambda_{\max}(H_t)$ and the gradient-noise covariance $\Sigma(\theta_{t-1})$ characterize local geometry and stochasticity, but can be expensive to estimate online. Their pointwise values also leave open how successive updates interact and accumulate over a training trajectory. This motivates complementary diagnostics that describe directional reinforcement and cancellation using quantities available during optimization.

D. Practical Stability Analysis in Large-Model Pretraining

Large-model training reports approach stability through observable symptoms and interventions on the training recipe. An instructive example is PaLM: Section 5.1 reports approximately 20 loss spikes during training of the 540B model despite global gradient clipping. The authors recovered by returning to a checkpoint about 100 steps before a spike and skipping 200–500 batches. Replaying those batches from a different parameter state did not reproduce the spikes, suggesting an interaction between data and model state rather than a property of the batches alone [24].

Reports accompanying released model weights provide further operational evidence. During OPT-175B training, loss divergence was associated with failures of the dynamic loss scale and growing final-layer activation norms. Recovery involved reducing the learning rate and restarting from an earlier checkpoint; lowering the gradient-clipping threshold from 1.0 to 0.3 also improved stability early in training [25]. These observations motivate monitoring the numerical state of training alongside the loss.

OLMo 2 makes this diagnostic process explicit by measuring a rolling spike score on training loss and gradient norm, and using controlled ablations to study initialization, normalization, and optimizer settings. Its revised recipe combines global gradient clipping with changes including query–key normalization, reordered normalization, and a logit regularizer (z-loss) [26]. In Llama 3, the authors report a stable pretraining recipe with few loss spikes and no interventions for divergence; they use learning-rate warmup and a staged batch-size schedule, and emphasize FP32 gradient accumulation and reduction for numerical stability [27]. These reports document both failure recovery and recipes that avoid recurrent failures, rather than a universal instability mechanism.

Loss, gradient norm, and activation statistics are practical monitoring signals, but gradient magnitude alone does not describe the relationship between successive directions. A large gradient can reinforce or oppose its predecessor, and a sequence of individually bounded steps can still exhibit cumulative cancellation. We therefore seek a complementary description that can be computed from training gradients and interpreted through their temporal geometry. The following section develops parameter displacement and consecutive-gradient cosine similarity for this purpose, connecting the latter to directional curvature and gradient noise without requiring online Hessian eigendecomposition.

III. Characterizing Constructive and Destructive Regimes

Loss spikes and oscillatory updates motivate practical diagnostics that can be computed from gradients already available during training. We develop these diagnostics by relating parameter displacement, gradient magnitudes and alignment,

curvature and sampling noise. The proposed quantities characterize constructive and destructive regimes, and inform possible interventions without the additional overhead of computing Hessians.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. As before, $\theta_t \in \mathbb{R}^p$ denotes the parameters after update t, and g_t is the mean mini-batch gradient evaluated at θ_{t-1}. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{7}$$

For a model trained from scratch, θ_0 represents the initialized model that has not yet learned from the training data. After T updates, the trained parameters θ_T encode what the model has learned within that training budget. The displacement Δ_T captures the net parameter change produced by training. Expanding the squared displacement separates gradient magnitudes from the interactions between different steps

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \end{aligned} \tag{8}$$

The terms $\|g_t\|^2$ are nonnegative and determine the squared lengths of individual SGD steps through $\eta^2\|g_t\|^2$. In full-batch GD, a larger gradient norm indicates a steeper local loss and, therefore, a larger first-order decrease along the negative gradient. The cross-step terms $g_u^\top g_t$ describe how these steps interact – positive inner products reinforce the displacement, whereas negative inner products cancel part of it. Large gradient norms can therefore coexist with cancellation along the training trajectory, and destructive interference (i.e. persistent $g_u^\top g_t < 0$) can limit displacement growth.

To isolate directional alignment from gradient magnitudes, define the cosine similarity between any two nonzero gradients and its consecutive-step form

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \quad 1 \le u < t \le T. \tag{9}$$

In this setup, the alignment between consecutive gradients is $\rho_t := \rho_{t-1,t}$, $t = 2, \ldots, T$. Substituting into Eq. (8) expresses the squared displacement entirely through gradient norms and directional alignment

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}. \end{aligned} \tag{10}$$

For fixed gradient norms, positive $\rho_{u,t}$ increases the squared net displacement, negative $\rho_{u,t}$ contributes cancellation, and orthogonal gradients have no pairwise contribution. In practice, ρ_t provides an online diagnostic of the interaction between successive updates: persistent positive or negative values characterize constructive or destructive regimes.

A. Directional Curvature and Consecutive Alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise in Eq. (4), $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$. Suppose the full objective $\mathcal{L}$ is twice continuously differentiable along the preceding step, and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD update yields

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{11}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\begin{aligned} \|g_{t-1}\|\|g_t\|\rho_t \approx \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}. \end{aligned} \tag{12}$$

Positive curvature reduces the component of the next gradient along the preceding direction. Its strength relative to the preceding squared norm is the directional curvature

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{13}$$

Thus, κ_t measures how strongly a step changes the gradient along the direction actually followed, whereas ρ_t measures the resulting alignment. Sampling modifies this response through the normalized noise interaction

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{14}$$

Dividing Eq. (12) by the two gradient norms gives

$$\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t). \tag{15}$$

In full-batch GD, $\varepsilon_t = 0$. Within this first-order approximation, $\eta\kappa_t < 1$ retains a positive projection, while $\eta\kappa_t > 1$ reverses it: the next update partially opposes the preceding step. At $\eta\kappa_t = 1$, the new gradient has lost its component along the old direction. It is orthogonal if nonzero; if it vanishes, cosine is undefined. The norm ratio determines how this projected change translates into the magnitude of ρ_t.

The threshold at one also has an optimization interpretation. Along the full-gradient direction, the local quadratic loss model is $\mathcal{L}(\theta_{t-2} - \alpha g_{t-1}) \approx \mathcal{L}(\theta_{t-2}) - \alpha\|g_{t-1}\|^2 + \alpha^2\kappa_t\|g_{t-1}\|^2/2$. For $\kappa_t > 0$, its directional minimum is at $\alpha = 1/\kappa_t$. Negative alignment therefore indicates that the step has passed this directional minimum under the approximation. Other gradient components may remain: this threshold identifies directional overshoot, rather than a stationary point or the loss-increase boundary derived below.

This interpretation leads to a measurable diagnostic. Rearranging Eq. (15) gives

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx \eta\kappa_t - \varepsilon_t. \tag{16}$$

The left-hand side uses only consecutive gradients: it measures their finite-step projected change without a Hessian evaluation. In full GD it estimates directional curvature when the Taylor remainder is small; in SGD it also contains sampling noise. It characterizes the realized step, whereas Eq. (3) concerns perturbations along the most curved local direction.

B. Noise Contribution under Mini-Batch Sampling

Unbiased sampling does not make the noise interaction vanish. Write $g_{t-1} = \nabla\mathcal{L}(\theta_{t-2}) + \xi_{t-1}$. Fresh, conditionally unbiased batches give $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. With finite second moments, the preceding gradient contributes its own noise energy:

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{17}$$

Thus, even unbiased sampling noise contributes a systematic nonpositive term to the expected unnormalized interaction between consecutive gradients.

This contribution is directly related to mini-batch size. Consider a mini-batch of b examples sampled uniformly without replacement from $N > 1$ training examples. Using the per-sample gradient covariance $\Sigma(\theta)$ defined in Eq. (5), the sampling variance satisfies

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{18}$$

At fixed parameters, smaller batches produce larger sampling variance and, consequently, a larger negative expected noise contribution. For full-batch gradients ($b = N$), this contribution vanishes.

Noise can weaken directional coherence by enlarging gradient norms relative to their shared signal. For independent estimates at two fixed states, let μ, ν be the nonzero full gradients and v_μ, v_ν the expected squared noise norms. When sampled norms concentrate,

$$\mathbb{E}[\cos(\mu+\xi, \nu+\zeta)] \approx \frac{\cos(\mu,\nu)}{\sqrt{(1+v_\mu/\|\mu\|^2)(1+v_\nu/\|\nu\|^2)}}.$$

This attenuation moves either sign toward zero. At the same fixed state, the common full-gradient direction gives the special case

$$\mathbb{E}[\rho_t] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{19}$$

Here the expected cosine is nonnegative; the approximation quantifies the dilution of a common direction by sampling. During SGD, a sampled update also changes the next state, coupling noise and curvature. The fixed-state attenuation argument therefore isolates a mechanism, while the batch-size intervention in Section IV-D tests its expression along evolving trajectories.

C. Normalized Displacement and Cumulative Interference

While the consecutive cosine similarity ρ_t measures local lag-1 directional alignment, it does not capture interactions between more distant updates (lag$\ge$ 2). To measure their cumulative effect, we define the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta\sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{20}$$

For constant-step plain SGD, the denominator represents the total distance traveled in parameter space, while the numerator measures the net displacement from initialization. Thus, R_T quantifies the fraction of accumulated step length retained as net displacement.

Using Eq. (10), this ratio satisfies

$$R_T^2 = \frac{\sum_{t=1}^T \|g_t\|^2 + 2\sum_{1\le u<t\le T}\|g_u\|\|g_t\|\rho_{u,t}}{\left(\sum_{t=1}^T \|g_t\|\right)^2}. \tag{21}$$

Unlike ρ_t, which considers only adjacent gradients, R_T incorporates interactions across all temporal lags. When R_T is near one, almost all traveled distance survives as net displacement; when it is near zero, little does. Repeated back-and-forth steps and weakly coherent sampled directions both suggest low retention. For example, equal-length orthogonal updates give $R_T = 1/\sqrt{T}$, whereas equal opposing pairs cancel exactly. We therefore expect small R_T when such behavior persists over a substantial part of an SGD path. Its value is relative to traveled distance: absolute displacement can still grow, and the sign of the complete interaction in Eq. (21) requires the full decomposition.

D. Directional Overshoot and Loss Increase

Directional overshoot need not undo the loss decrease already made along a step. Define its full-objective change as $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & + \underbrace{\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t)}_{\text{Noise interaction}}.\end{aligned} \tag{22}$$

The magnitude term favors descent. The interference term reinforces it for positive alignment and opposes it for negative alignment: destructive interference contributes positively to the predicted loss change. With negligible noise and nonzero gradients, the approximation predicts a loss increase precisely when

$$\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}. \tag{23}$$

This requires both negative alignment and gradient-norm growth. Neither alone suffices: the opposing component of the new gradient must exceed the preceding gradient magnitude. Large norms can therefore accompany descent, just as negative cosine can accompany a smaller loss decrease.

Substituting Eq. (15) with zero noise gives

$$\delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{24}$$

The two thresholds distinguish overshooting the directional minimum ($\eta\kappa_t > 1$) from overshooting far enough to raise the loss ($\eta\kappa_t > 2$). Between them, the gradients oppose each other while the step still improves the objective. These are local quadratic predictions; finite-step nonlinear effects and the explicit SGD noise term can change the observed sign. A transient increase becomes a loss peak only if subsequent updates recover.

IV. Empirical Characterization of Gradient Interference

We use full-batch GD to characterize directional overshoot, its effect on cumulative motion, and its coexistence with loss reduction. We then vary batch size to examine how sampling changes these signatures.

The comparison covers a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its complete fixed training objective for 3,000 updates at three constant learning rates (Table I). Within each model, rates share one frozen initialization and data preparation; they were selected by earlier training calibrations to expose contrasting dynamics, without test-loss selection. Plain GD uses no momentum, weight decay, clipping or schedule. These comparisons test whether the same geometric signatures recur across the three objectives; they do not isolate architecture effects. Appendix C specifies objectives, architectures, precision and measurement details.

A. A Gradient-Based View of Oscillatory Descent

Figure 1 pairs loss with the projected gradient change $\eta\widehat{\kappa}_t$. The higher-rate trajectories develop oscillatory loss while continuing to improve over longer timescales; their late projected change approaches two. Equation (24) explains why this value is informative: in the local quadratic model, it separates a loss-decreasing step from one that reverses the preceding progress. The observed pattern gives a gradient-based view of the oscillatory descent associated with the Edge of Stability [13].

The diagnostic requires consecutive gradients rather than Hessian–vector products or an eigensolver. It estimates curvature along the realized step when the local approximation applies; it does not estimate the largest Hessian eigenvalue that defines spectral EoS. This distinction permits a practical interpretation without treating the two quantities as interchangeable. The same local model explains the regime but does not reliably predict every nonlinear loss fluctuation: the stepwise approximation check and spectral coverage are retained in Appendix C.

B. Reversal, Step Magnitude and Displacement Retention

Figure 2 shows how the oscillatory regime changes parameter motion. Persistent negative cosine means that successive updates repeatedly oppose the preceding direction, consistent with overshooting its local directional minimum. The lowest MLP and Transformer rates retain positive alignment, whereas the other seven conditions approach nearly opposing gradients. In the MLP and ResNet, the transition also accompanies growing gradient norms: the steps become larger even as their directions increasingly cancel. These fluctuations arise without mini-batch sampling.

The decreasing R_t shows that larger steps need not translate into proportionally larger net motion. Repeated reversals consume path length, so the retained fraction can become small even while parameters continue to move away from initialization. This is a geometric inefficiency, not evidence of slower optimization. Table I makes the distinction concrete: within every model, the highest tested rate has the lowest retention, the largest absolute displacement, and the lowest final training loss. A rate can therefore be large relative to local directional minima yet productive over the full training horizon.

Adjacent cancellation also does not determine the sign of the complete cross-step interaction. That total is positive at the endpoint in all nine trajectories, including the seven with negative accumulated lag-one interaction (Table III). Longer-range reinforcement can coexist with repeated local reversal. Together, ρ_t, gradient norms and R_t distinguish local overshoot, step size and retained motion; loss establishes whether this motion improves the objective.

C. Oscillation Beyond Adjacent Updates

To examine the temporal extent of reversal, we fix one endpoint gradient and compare it with preceding gradients at every lag $k = 1, \ldots, 256$. Figure 3 uses the largest tested rate for each model and the common endpoint after 2,256 updates. Near-zero similarity at some lags does not imply that directional memory has disappeared: strongly opposing and reinforcing pairs can reappear at longer separations. The varying amplitude and sign pattern show that oscillation extends beyond a single adjacent pair, while its persistence depends on the model and part of the trajectory. This temporal structure explains why a lag-one statistic cannot summarize the complete interference term.

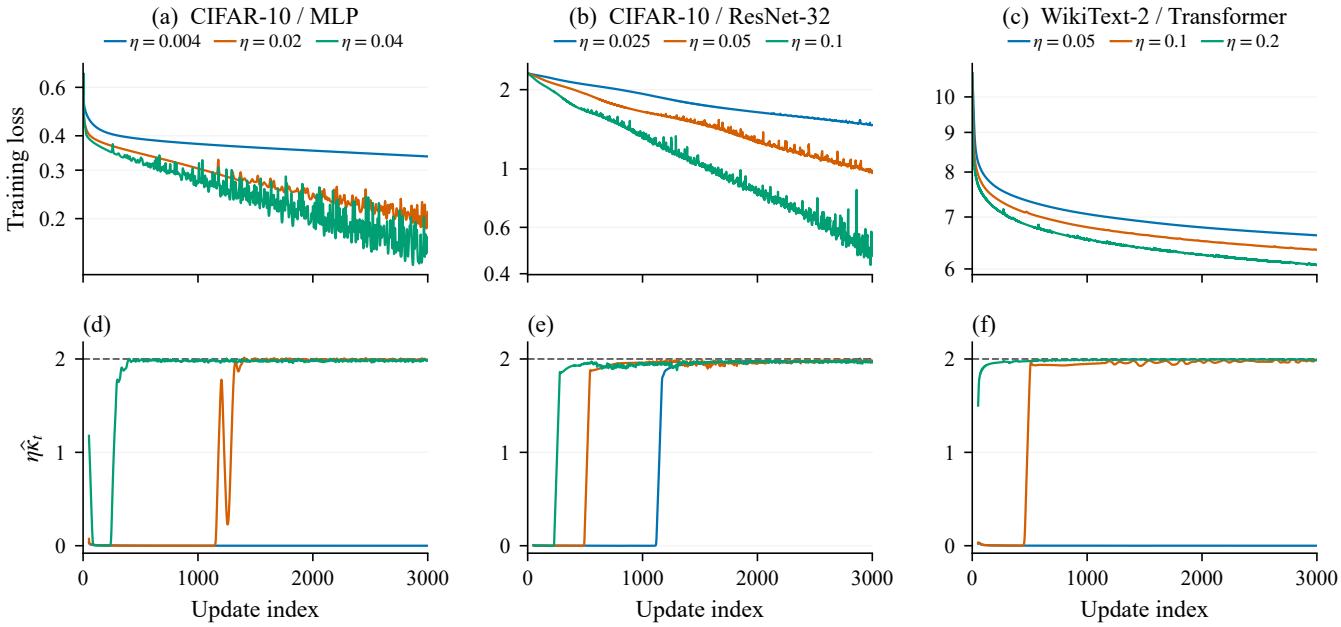


Fig. 1: Training loss and projected gradient change under full-batch GD. Columns identify the dataset/model; colors identify learning rates within each column. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: complete 50-pair trailing means of the pointwise $\eta\widehat{\kappa}_t$, with the local quadratic loss-change boundary at two. Loss at index t is evaluated at θ_t; gradient diagnostics use g_t at θ_{t-1}. Each model uses one shared initialization. Smoothing summarizes the regime and does not identify its exact onset.

TABLE I: Endpoints at $T = 3000$ updates. The late means $\overline{\rho}_{\text{late}}$ and $\overline{\eta\widehat{\kappa}}_{\text{late}}$ average gradient indices $t = 2001, \ldots, 3000$. Displacement is $\|\Delta_T\| = \eta\|\sum_{t=1}^{T} g_t\|$, subject to parameter-update roundoff. Each model shares one initialization across rates; comparisons are within models. *Positive value below 0.00005, rounded to zero.

Dataset	Model	η	$\mathcal{L}(\theta_T)$	$\overline{\rho}_{\text{late}}$	R_T	$\overline{\eta\widehat{\kappa}}_{\text{late}}$	$\|\Delta_T\|$
CIFAR-10	MLP	0.004	0.3365	1.0000	0.7941	0.0000*	0.9511
		0.02	0.2114	-0.9913	0.0488	1.9924	3.1391
		0.04	0.1539	-0.9799	0.0257	1.9820	4.7628
	ResNet-32	0.025	1.4665	-0.9723	0.1385	1.9740	5.7045
		0.05	0.9672	-0.9708	0.0616	1.9746	9.0183
		0.1	0.4898	-0.9684	0.0374	1.9707	13.9329
WikiText-2	Transformer	0.05	6.6301	1.0000	0.7223	0.0002	11.3183
		0.1	6.3519	-0.9749	0.1395	1.9754	15.6026
		0.2	6.0697	-0.9937	0.0541	1.9937	21.9693

D. Mini-Batch Noise and Directional Coherence

We vary batch size at initialization and after 2,000 GD updates in the same three model–dataset cases, holding $\eta = 0.02$, 0.1 and 0.2 fixed for the MLP, ResNet-32 and Transformer, respectively. Each starting state receives 200-update continuations, with three initializations, eight sampling repetitions per stochastic condition and one full-GD control per initialization. Figure 4 compares mean sampled consecutive-gradient cosine over each continuation. Equal update budgets imply unequal data exposure; sampling units, objective coverage and numerical checks are in Appendix C.

The smallest batches weaken trained-state reversal in all three cases. Initial-state responses are less uniform: full GD has positive mean cosine in the MLP and ResNet and strongly negative mean cosine in the Transformer, where reversal develops within the first 200 updates. The MLP's sampled paths can also acquire negative mean alignment while its full-GD mean is positive. Sampling therefore changes the regime encountered along a path, in addition to perturbing gradients at fixed parameter states.

For the Transformer at $b = 8$, both starting states have mean cosine near -0.26, yet mean full-data loss falls from initialization and rises after GD training. Mean adjacent alignment alone therefore does not order training-objective progress.

The MLP's smallest trained-state batch likewise raises mean full-data loss, whereas full GD lowers it in both models.

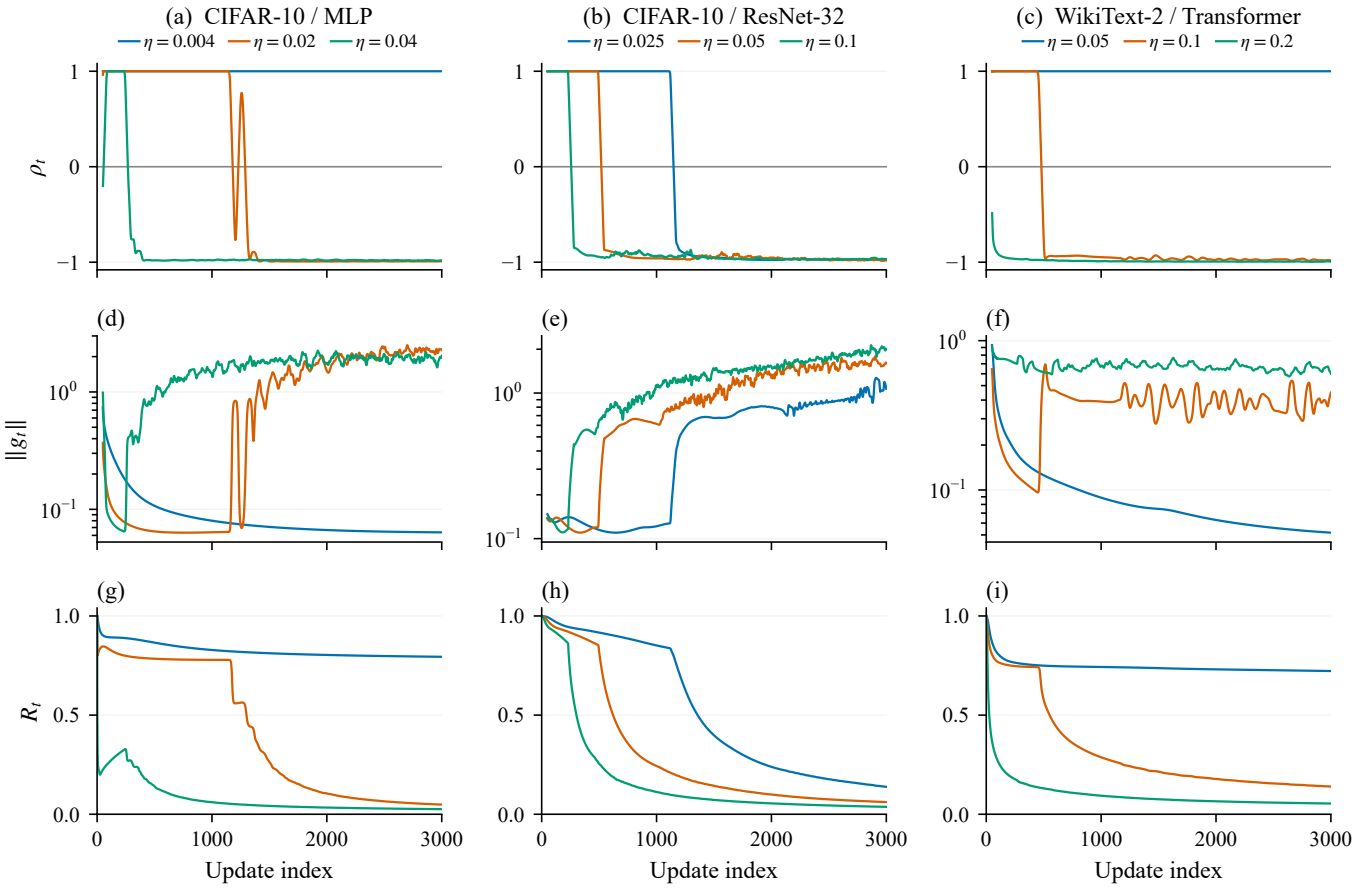


Fig. 2: Directional coherence and cumulative motion for the conditions in Fig. 1. Rows show consecutive cosine, gradient norm (logarithmic scale), and normalized displacement. Cosines and norms use complete 50-observation trailing means; R_t is the unsmoothed ratio over the complete prefix through update t. Colors and column ordering match Fig. 1. Raw cosine traces appear in Appendix C.

Noise covariance can affect the solutions favored by training [28], but mean cosine alone identifies neither that bias nor a generalization benefit. These comparisons describe selected objectives, rates and update budgets; they do not select an optimal batch size.

V. Raw-Gradient Diagnostics under Adam

In Adam, raw-gradient dynamics provide information about impending loss excursions beyond recent loss history, even though Adam transforms those gradients before updating the parameters. The raw gradient observes the loss response along the adaptive path; the first moment records how that signal is averaged over time. Both are already computed by the optimizer.

For Adam [16],

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{25}$$

with zero initial moments. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, and

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{26}$$

where operations are coordinatewise and $\epsilon > 0$ is a numerical constant. Norms and consecutive cosines of g_t and m_t require vector reductions and a previous-vector buffer, without another backward pass.

The projected-change statistic in Eq. (16) also remains available:

$$\eta\widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}. \tag{27}$$

Under Adam it measures the gradient response along the adaptive trajectory, rather than SGD directional curvature. For nonzero gradients, a value above two requires negative cosine and gradient-norm growth; these are coupled observations, not independent warning signals. Momentum supplies a different view: its norm and cosine show how that changing input propagates through temporal averaging.

A. Controlled Full-Batch Comparison

We train the same three objectives from their frozen GD initializations, with fresh Adam moments and 1,024 full-objective updates at each constant rate. The MLP uses $\eta \in \{0.0001, 0.0003, 0.001\}$; ResNet-32 and the Transformer use $\{0.0003, 0.001, 0.003\}$. All arms use $(\beta_1, \beta_2) = (0.9, 0.999)$

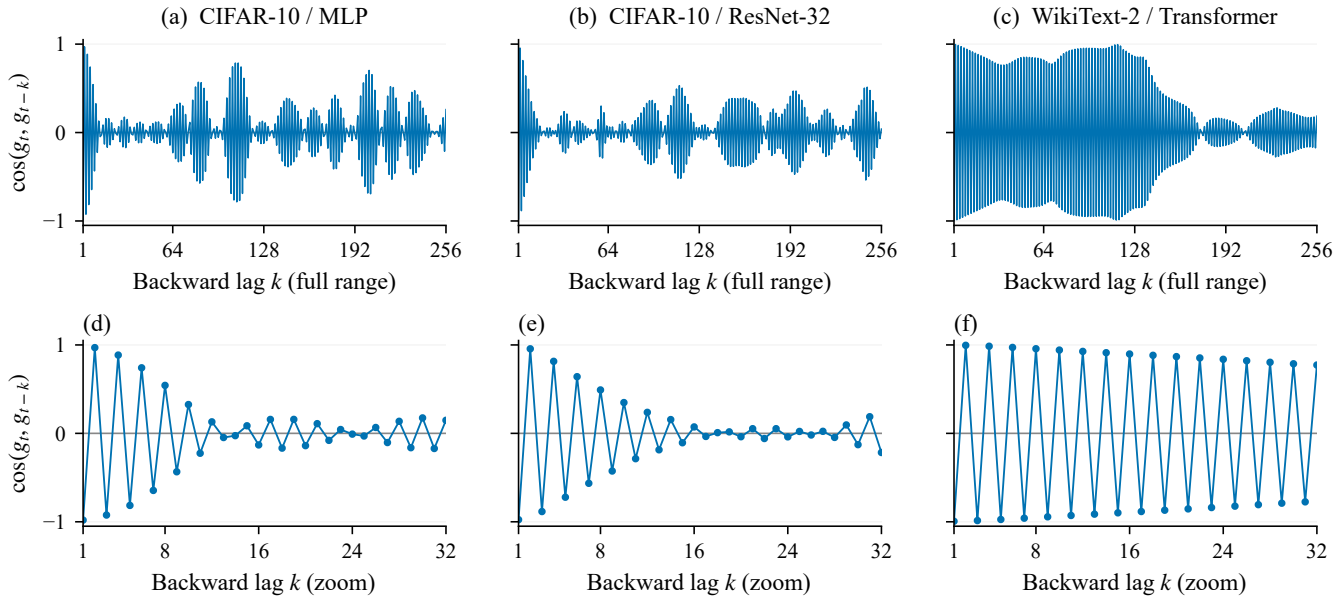


Fig. 3: Backward gradient alignment at the common endpoint state θ_{2256}, corresponding to $t = 2257$ under the manuscript's preupdate gradient indexing. Each point is one exact cosine $\cos(g_t, g_{t-k})$; the endpoint is fixed across all 256 lags. The top row shows lags 1–256; the bottom row magnifies the same values at lags 1–32 to resolve the early sign alternation. Rates are 0.04 (MLP), 0.1 (ResNet-32) and 0.2 (Transformer). These are separately recomputed 256-update GD continuations from retained state 2000, with one initialization and no smoothing or endpoint averaging. Appendix C records the additional endpoints and numerical replay scope.

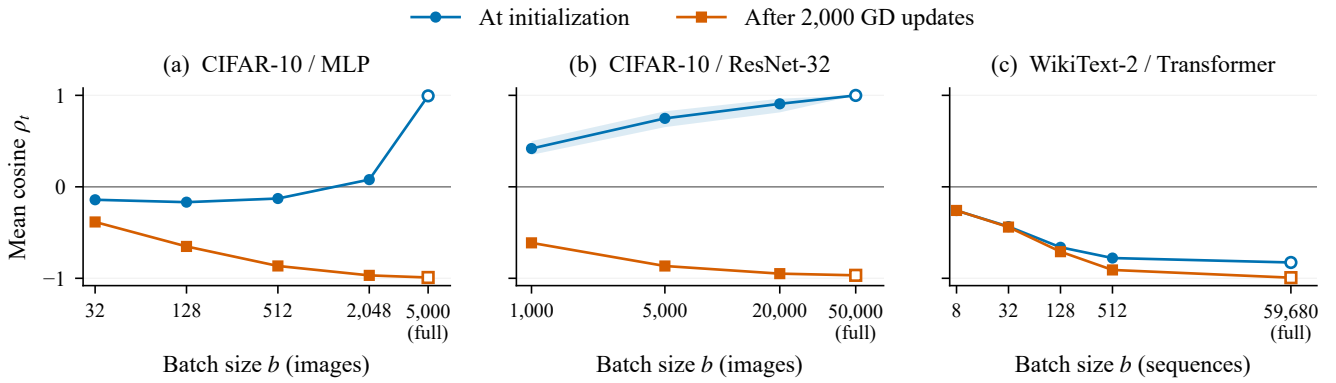


Fig. 4: Mean sampled-gradient cosine over 199 adjacent pairs during 200-update continuations in three model–dataset cases. Blue circles start at initialization; orange squares start after 2,000 GD updates. Each line averages repetitions within initialization, then weights the three initialization blocks equally; shading is their range, not a confidence interval. Hollow rightmost markers denote full GD. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic; lines guide the eye between measured sizes.

and $\epsilon = 10^{-8}$. There is one initialization per case. Objectives, numerical checks and the exploratory spike-analysis protocol are detailed in Appendix C-G.

Figure 5 separates local gradient reversal from what temporal averaging retains. The recorded adaptive directions show a further distinction: at the Transformer's smallest rate, the last 64 updates have median raw cosine -0.795 but adaptive-direction cosine 0.939: Adam can move coherently while successive raw gradients oppose one another. Conversely, the MLP eventually exhibits reversal in both raw gradients and momentum. Temporal averaging therefore modifies the interference pattern without guaranteeing a positive first-moment cosine.

The largest tested rate attains the lowest final training loss in each case, despite visible intermediate spikes. These excursions can be transient features of an improving trajectory. Neither negative raw cosine nor a spike by itself orders the attained loss across rates.

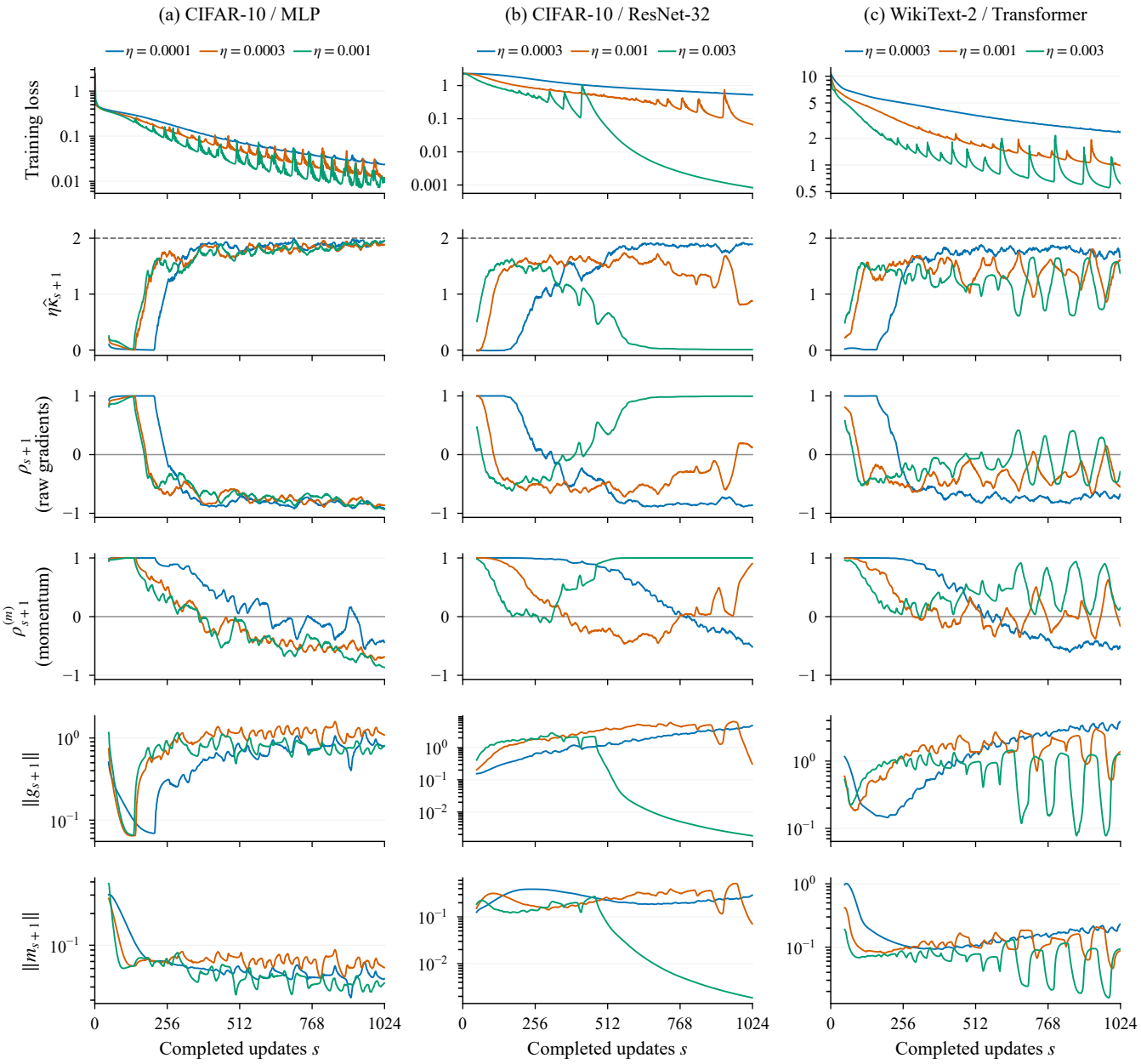


Fig. 5: Controlled Adam training at three constant rates per case. Training loss is unsmoothed; the other rows use complete 50-observation trailing means. All rows are aligned to parameter state θ_s: g_{s+1} and m_{s+1} are available before the next update. Momentum denotes the stored first moment, without bias correction. The dashed reference at two concerns raw-gradient projected change, not an identified Adam stability boundary. One initialization per case.

B. Information beyond Recent Loss

We test whether the diagnostics add information before a pronounced excursion, rather than merely accompany its peak. From states still within 5% of their recent loss minimum, we forecast a rise of at least 10% in the next eight updates. A baseline using recent loss, training age and task/rate indicators is compared with the same model plus continuous raw-gradient projected change. Training uses expanding prefixes; evaluation uses five later blocks separated from fitting by a 24-state gap. Appendix C-G gives the complete protocol.

Table II shows additional information in every case. Pooled average precision increases from 0.260 to 0.454, with improvement in all five time blocks. Thus, the gradient response distinguishes future excursions that recent loss and the known training context alone predict less effectively. This is exploratory temporal evidence from the controlled full-batch trajectories, with one initialization per case. The raw views below explain the timing of these signals.

The raw views in Fig. 6 resolve changes hidden by the overview's averaging. In the Transformer, projected change

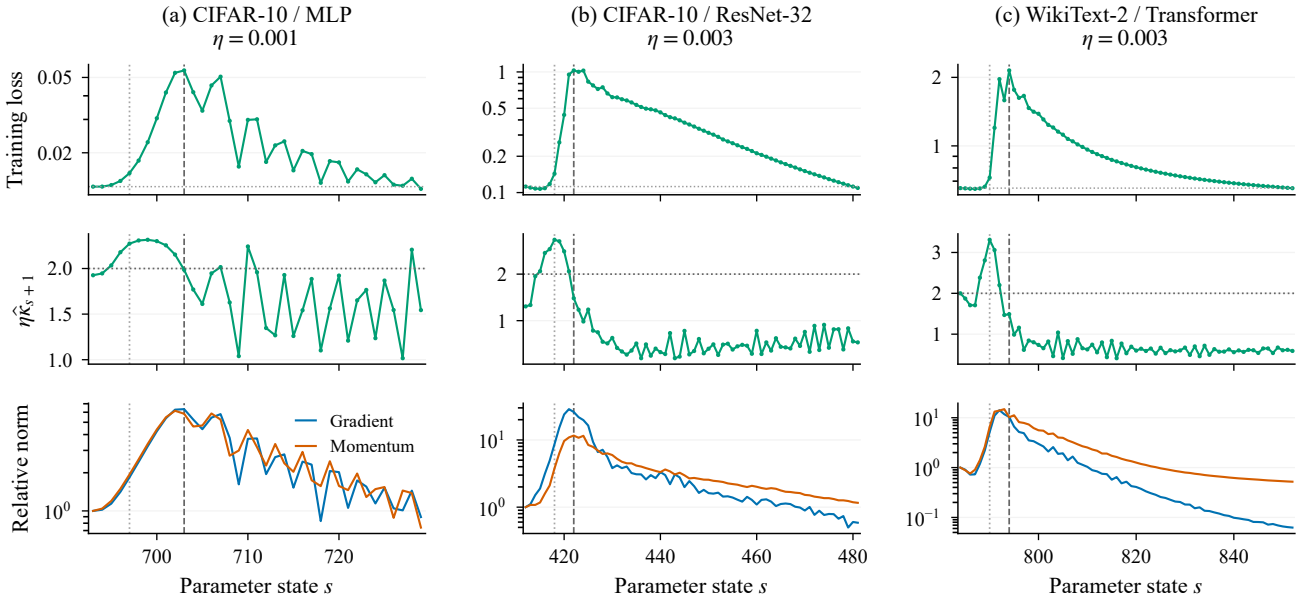


Fig. 6: Unsmoothed views of the completed loss episode with the largest peak/baseline ratio at each case's highest tested Adam rate. Each window begins ten states before its peak and ends at the first subsequent loss below the starting value. Vertical dotted lines mark the episode's 10% rise onset; dashed lines mark the peak. Norms are divided by their own value at the window's start. Gradient and momentum diagnostics use the same state clock as Fig. 5. These are loss-selected examples; repeated events and non-spike states are included in the quantitative analysis.

TABLE II: Average precision for impending loss excursions on the same forward-test states. The last column adds raw-gradient projected change to the loss-history baseline; task/rate and training age enter both. Positive prevalence is the fraction of states with a subsequent rise. Per-case rows evaluate the pooled models within each case.

Case	Positive (%)	Loss history	+ raw change
MLP	28.7	0.500	0.546
ResNet-32	3.8	0.045	0.136
Transformer	6.6	0.248	0.433
Pooled	10.9	0.260	0.454

crosses two at state 788, when loss is only 0.40% above its recent trough; loss reaches a 10% rise at 790 and peaks at 794. Gradient and momentum norms grow sharply at the crossing and peak at 792 and 793, respectively. The signal precedes the pronounced excursion, although the first small loss increase has already occurred. Similar ordering appears in the selected MLP and ResNet episodes, with different delays.

To assess repetition, we identify episodes from loss alone: a rise of at least 10% above the preceding 16-state minimum, followed until recovery below that fixed baseline. Among episodes reaching at least 50%, a fresh crossing of two occurs 1–8 states before onset in 13/17 MLP, 3/3 ResNet and 7/7 Transformer episodes at the highest rates, while loss is still within 5% of its recent minimum. These are recurrent events within one trajectory per rate, rather than independent initialization replicates.

The threshold also fires without a spike. Across five forward test blocks covering all nine arms, only 187 of 891 eligible states above two are followed by a 10% loss rise within eight updates (21% precision). Both the lowest-rate ResNet and Transformer exhibit threshold crossings without a qualifying episode. The incremental information in Table II therefore supports a continuous diagnostic, while a universal threshold discards important context.

Raw-gradient dynamics remain informative even when adaptive updates differ substantially from the gradients themselves. Momentum helps interpret how these changes enter the optimizer's memory. Their joint growth near spikes is compatible with amplification, but higher loss need not imply a larger gradient norm, so temporal ordering alone does not establish that feedback mechanism. A useful scheduling rule must further demonstrate that acting on the diagnostic improves subsequent loss beyond decisions based on recent loss alone.

VI. Conclusion

The displacement formulation distinguishes the magnitude of individual updates, their local directional interaction and their cumulative accumulation. For plain SGD, the exact decomposition retains all temporal lags, while consecutive cosine provides a tractable local observation. The curvature and sampling relations clarify how that observation can change without determining either cumulative displacement or loss improvement. The full-batch and mini-batch experiments illustrate these distinctions: gradient reversal can coexist with objective progress, and weaker reversal need not improve displacement retention or the attained loss.

In Adam, raw-gradient dynamics provide information about impending loss excursions beyond recent loss history, even though the optimizer transforms those gradients before updating the parameters. The controlled full-batch comparisons support this interpretation through forward-time prediction and the timing of gradient/momentum changes around spikes. Local reversal can coexist with coherent adaptive directions and continued objective progress. Frequent threshold false alarms distinguish the information in the continuous diagnostic from a universal spike trigger.

The practical contribution is a framework for interpreting these measurements together and using them to formulate controlled interventions. Developing a scheduler requires a further step: demonstrating that acting on the diagnostics improves loss beyond decisions based on recent loss and known training settings, at matched budgets and across independent runs. That intervention evaluation remains future work.

APPENDIX A
DERIVATION OF THE ALIGNMENT-BASED LOSS CHANGE

Consider the SGD update $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$. The change in the full objective along this step can be expressed exactly as

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds.\end{aligned} \tag{28}$$

Approximating the integral by the average of the full gradients at the two endpoints gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top \left(\nabla\mathcal{L}(\theta_{t-2}) + \nabla\mathcal{L}(\theta_{t-1})\right).\end{aligned} \tag{29}$$

This approximation is exact for quadratic objectives. For nonlinear objectives, the approximation error is of third order in the step length when the Hessian is locally Lipschitz continuous.

Using the stochastic gradient decomposition $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$ at both endpoints, we obtain

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top (g_{t-1} + g_t - \xi_{t-1} - \xi_t) \\ &\quad = -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t).\end{aligned} \tag{30}$$

Substituting $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ and factoring the preceding squared gradient norm yields

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2}\|g_{t-1}\|^2 \left(1 + \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}\right) \\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t),\end{aligned} \tag{31}$$

which recovers Eq. (22).

To connect this expression with directional curvature, recall from Eq. (15) that

$$\rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx 1 - \eta\kappa_t + \varepsilon_t. \tag{32}$$

Substituting this relation and using the definition of ε_t in Eq. (14) gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\eta\|g_{t-1}\|^2 \left(1 - \frac{\eta\kappa_t}{2}\right) + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{33}$$

Thus, the curvature-based and alignment-based expressions describe the same local loss change to second order. Positive directional curvature counteracts the first-order loss decrease, while its effect is already incorporated into the weighted alignment between consecutive gradients. The noise contributions account for the difference between the observed mini-batch gradients and the full gradients.

APPENDIX B
DERIVATION OF THE LOCAL LOSS CHANGES

We use the same indexing as Section III: g_{t-1} is evaluated at θ_{t-2}, g_t at θ_{t-1}, and $H_{t-1} = \nabla^2\mathcal{L}(\theta_{t-2})$. The loss is assumed twice continuously differentiable along the displacements considered below. The relations retain second-order terms and omit the Taylor remainders; they are exact when the objective is quadratic.

A. One Update and Directional Curvature

For $\theta_{t-1} - \theta_{t-2} = -\eta g_{t-1}$, Taylor expansion gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\eta\nabla\mathcal{L}(\theta_{t-2})^\top g_{t-1} + \frac{\eta^2}{2} g_{t-1}^\top H_{t-1} g_{t-1}.\end{aligned} \tag{34}$$

Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $g_{t-1}^\top H_{t-1} g_{t-1} = \kappa_t\|g_{t-1}\|^2$ yields Eq. (22) for nonzero g_{t-1}.

The positive term $\eta\xi_{t-1}^\top g_{t-1}$ in that expression cannot be interpreted separately as the net effect of noise. Conditional on the parameters before the batch is drawn, unbiased sampling gives

$$\begin{aligned}\mathbb{E}[\|g_{t-1}\|^2 \mid \theta_{t-2}] &= \|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ &\quad + \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}], \\ \mathbb{E}[\xi_{t-1}^\top g_{t-1} \mid \theta_{t-2}] &= \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}].\end{aligned} \tag{35}$$

Thus, the contributions from noise variance cancel between the first-order terms when taking their conditional expectation. With finite moments for the retained terms, the expected loss change in the quadratic approximation is

$$\begin{aligned}&\mathbb{E}[\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \mid \theta_{t-2}] \\ &\quad \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ &\qquad + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{36}$$

Sampling still contributes to the curvature-weighted second-order term. This conditional calculation does not remove that contribution or the Taylor remainder from a nonlinear trajectory.

B. Two Updates and Consecutive Alignment

The displacement over two SGD updates is exactly

$$\theta_t - \theta_{t-2} = -\eta(g_{t-1} + g_t).$$

Expanding around the starting parameters θ_{t-2} gives

$$\begin{aligned}\mathcal{L}(\theta_t) - \mathcal{L}(\theta_{t-2}) \\ \approx -\eta \nabla \mathcal{L}(\theta_{t-2})^\top (g_{t-1} + g_t) \\ + \frac{\eta^2}{2}(g_{t-1} + g_t)^\top H_{t-1}(g_{t-1} + g_t).\end{aligned} \tag{37}$$

The first-order projection can be written as

$$\begin{aligned}\nabla \mathcal{L}(\theta_{t-2})^\top (g_{t-1} + g_t) \\ = \|g_{t-1}\|^2 + g_{t-1}^\top g_t - \xi_{t-1}^\top (g_{t-1} + g_t).\end{aligned} \tag{38}$$

Finally, $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ yields a form of Eq. (37) using consecutive-gradient cosine. The same starting Hessian appears throughout this expansion. For a nonlinear objective, its remainder is governed by the combined displacement; a small cosine alone does not control that approximation error.

Appendix C
Measurement Details and Additional Diagnostics

A. Objectives and Measurement

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten output coordinates and averaged over the first 5,000 training images. ResNet-32 minimizes mean cross-entropy on all 50,000 training images. The word-level Transformer has two layers, two attention heads, and embedding/feed-forward dimensions of 200; its objective is token-weighted negative log-likelihood. Learning rates are constant and share frozen seed-0 initial tensors within each case. There is no momentum, decay, clipping, augmentation, dropout or schedule.

The MLP uses coordinate-wise standardization and ResNet uses channel-wise standardization, both based on the complete CIFAR-10 training split. ResNet's objective averages the same 50 fixed groups of 1,000 images, with batch normalization performed separately within each group. Accumulating all group gradients gives the gradient of this fixed objective; physical batching introduces no random sampling.

For the Transformer, the corpus is arranged into 20 fixed streams and causal chunks of at most 35 tokens. Eight stream-tail tokens and the 20 initial context tokens have no training target; short chunks retain their valid targets, and padding is excluded from the loss. The resulting 59,680 sequences contain 2,088,600 training targets. The 33,278-word vocabulary follows the first-occurrence dictionary over training, validation and test text in the PyTorch word-language-model example, revision `13acec6d7c78` [29]. The same pinned example supplies the Transformer implementation. These comparisons concern training dynamics rather than held-out generalization.

Physical accumulation uses 1,000-image chunks for the MLP, the fixed 1,000-image groups for ResNet, and 64-sequence chunks for the Transformer. The complete objective is accumulated before each parameter update. Training parameters and gradients are FP32, TF32 is disabled, and diagnostic reductions are FP64. The nine conditions are paired learning-rate comparisons within three model cases, not nine independent initializations or a controlled comparison of architectures. The gradient diagnostics require vector reductions and retained gradient information rather than online eigendecomposition; their runtime and memory overhead have not been isolated by a controlled benchmark.

The first adjacent pair is omitted without a denominator offset. The complete 50-observation trailing means aid visualization and do not define regime-onset times. Endpoint losses and R_{3000} are unsmoothed.

B. Unsmoothed Alignment and Temporal Accounting

Figure 7 supplies the unsmoothed consecutive cosines underlying Fig. 2(a)–(c). Complete trailing means suppress short transients and shift the displayed transition relative to the first negative pair. In particular, the MLP at $\eta = 0.04$ reverses near initialization and then returns to positive alignment before its later negative phase. These raw traces distinguish such a transient from a persistent pattern without introducing a stability threshold or discarding early observations.

For the complete prefix of T gradients, define the traveled distance and the temporal contributions as

$$\begin{aligned}P_T &:= \eta \sum_{t=1}^{T} \|g_t\|, \\ I_T^{(1)} &:= 2 \sum_{t=2}^{T} g_{t-1}^\top g_t, \\ I_T^{(>1)} &:= 2 \sum_{\substack{1 \le u < t \le T \\ t-u > 1}} g_u^\top g_t, \\ I_T &:= I_T^{(1)} + I_T^{(>1)} = \Big\| \sum_{t=1}^{T} g_t \Big\|^2 - \sum_{t=1}^{T} \|g_t\|^2.\end{aligned} \tag{39}$$

Table III reports these quantities at $T = 3000$. The interaction columns use raw gradients and include the factor of two; they are not multiplied by η^2. The total is positive in all nine conditions, whereas the adjacent contribution is negative in seven. This endpoint result does not assert positivity at every prefix or for every individual pair. Comparisons of magnitudes are within each model case and its fixed parameterization.

C. Retrospective Loss Approximation

For each gradient pair indexed by $t = 2, \ldots, 3000$, let $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. We compare this observed change with the gradient-based approximation

$$\widehat{\delta\mathcal{L}}_{t-1} := -\frac{\eta}{2}\|g_{t-1}\|^2 \left(2 - \eta\widehat{\kappa}_t\right). \tag{40}$$

The normalized absolute discrepancy is

$$e_t := \frac{|\delta\mathcal{L}_{t-1} - \widehat{\delta\mathcal{L}}_{t-1}|}{\eta\|g_{t-1}\|^2}. \tag{41}$$

The denominator is the magnitude of the first-order full-GD descent term. It is nonzero for every measured pair. Table IV reports the median over all 2,999 pairs and the count of opposite signs. The final update is excluded because its subsequent

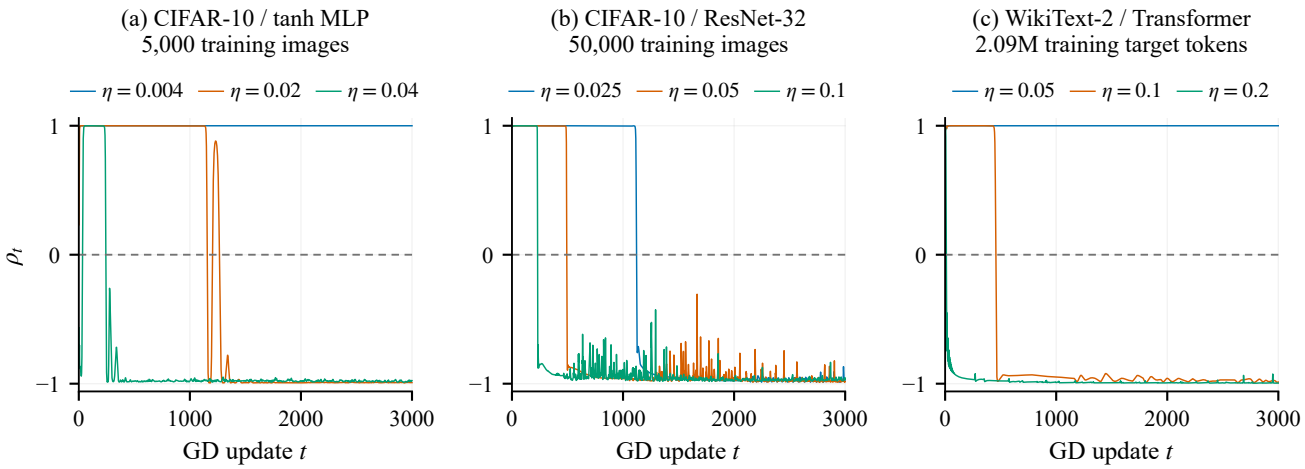


Fig. 7: Unsmoothed consecutive-gradient cosine for the nine full-GD conditions in Section IV, with 2,999 defined pairs per condition. Columns show (a) MLP, (b) ResNet-32 and (c) Transformer. The dashed horizontal reference is zero. Blue, orange and green identify increasing learning rates within each case, as given in the legends. Every curve uses seed 0 and covers the same 3,000-update horizon; the first undefined pair is omitted and there is no temporal or seed averaging. Coincident positive curves can overlap exactly.

gradient was not recorded. All inputs are unsmoothed; the table uses no seed aggregation.

For the threshold sensitivity columns, define

$$h_t(\tau) := \tau \max\{|\mathcal{L}(\theta_{t-2})|, |\mathcal{L}(\theta_{t-1})|\}. \tag{42}$$

A transition is eligible only when both $|\delta\mathcal{L}_{t-1}| > h_t(\tau)$ and $|\widehat{\delta\mathcal{L}}_{t-1}| > h_t(\tau)$. Each cell prints opposite-sign transitions divided by its eligible count; the median discrepancy remains computed over all pairs. These thresholds are descriptive sensitivity choices, not calibrated FP32 error bounds. Serial transitions are not independent initialization replicates, and the comparison is retrospective because it uses the subsequent gradient. The discrepancy does not isolate finite-step, nonsmooth or numerical effects. For the highest-rate Transformer, the unthresholded signs disagree in 44.2% of transitions; excluding changes below the 10^{-5} relative threshold leaves 43.7% disagreement. Thus a projected change near two is informative about the oscillatory regime but does not by itself validate a pointwise nonlinear loss prediction.

D. Backward-Lag Measurements

The backward profiles use the largest GD rate in each model, with exact full-objective gradients from separately recomputed 256-update continuations. Retained states 500, 2000 and 3000 supply common starting parameters; their endpoints are states 756, 2256 and 3256 (gradient indices 757, 2257 and 3257). Figure 3 uses the middle endpoint in every model, a common trained-state comparison rather than an endpoint chosen for maximum oscillation. Each curve holds that endpoint fixed, includes every lag through 256, and uses no temporal averaging or compression. FP64 products of FP32 vectors were checked independently on CPU and GPU; selected retained vectors supply additional local checks. Later floating-point trajectories can depart from earlier replays despite identical starting tensors, so these profiles are separate continuations, not reconstructed pairs from Figs. 1 and 2. At all nine measured endpoints, both positive and negative similarities occur within lags 193–256; their amplitudes vary substantially across endpoints.

E. Mini-Batch Comparison

The mini-batch comparison uses three initializations for each objective. The MLP reuses the completed 5,000-image experiment at $\eta = 0.02$, with batches $32, 128, 512, 2048, 5000$. Fresh 2,000-update GD references at $\eta = 0.1$ and 0.2 supply ResNet and Transformer starting states. ResNet samples $1, 5, 20$ or all 50 fixed 1,000-image normalization groups, preserving each image's normalization context; its figure axis reports image counts. Transformer batches contain $8, 32, 128, 512$ or all $59{,}680$ prepared sequences, with $M = 2{,}088{,}600$ supervised tokens in the complete training objective. If ℓ_i is sequence i's summed token loss, the sampled estimator is $\frac{N}{bM}\sum_{i\in B}\ell_i$, with $N = 59{,}680$; a random sampled-token denominator would change this estimator. Every stochastic batch is a uniform subset without replacement, independently redrawn at each update. Batch sizes share prefixes of a common permutation within a repetition; permutations are independent across updates and repetitions. Parameters and gradients use FP32, reductions use FP64, and TF32 is disabled. Full-loss evaluations cover every training image or supervised token. Cosines average all 199 defined consecutive pairs; retention uses all 200 applied gradients. Eight repetitions are averaged within initialization before the three initializations receive equal weight. The plotted range represents variation among initialization-block means, not sampling confidence. Cases differ in objective, architecture, data size and selected learning rate; their contrasts do not isolate any one factor.

Cloud runs use PyTorch 2.8/CUDA 12.8. The Transformer uses RTX5090 and 512-sequence physical accumulation, with-

out changing effective batches. ResNet uses L40S/RTX4090; seed 2's smallest two batches use GB10/PyTorch 2.11/CUDA 13 from hash-identical cloud checkpoints. Complete-gradient and short replay checks qualify this runtime; a 200-update initial full-GD bridge has endpoint parameter relative error 2.36×10^{-7} versus the cloud. This does not establish long-trajectory replay. Batch and runtime effects are confounded in this block; shading includes initialization and runtime variation. Initial full controls for ResNet seed 2 and Transformer seed 0 reuse the exact first 200 reference updates, with separate endpoint losses; other controls execute independently. Audits of actual increments and endpoint displacement quantify FP32 update roundoff.

F. Spectral Context

We measure the largest algebraic Hessian eigenvalue at states $0, 100, \ldots, 3000$ using matrix-free Hessian–vector products and a checked relative eigenpair residual of at most 0.005. The Hessian objective covers all 5,000 MLP training examples, but only the first five fixed ResNet groups and the first 2,500 Transformer sequences. Subset measurements follow the computational motivation of [13]; they are not full-objective sharpness for the larger models. Reused MLP trajectories requested two largest eigenvalues and the other runs requested one; we compare only the largest value. Table V retains all nine final measurements.

Maximum pointwise eigenvalue concerns the most curved local direction, whereas $\widehat{\kappa}_t$ concerns realized gradient change across a step. Objective coverage and a finite-step Taylor remainder also affect their comparison. Final spectral values and the late means in Table I use different temporal reductions. Their agreement or disagreement therefore does not isolate estimator accuracy or the cause of a discrepancy. A state-matched local comparison would pair the Hessian at state s with the expansion for gradient index $t = s + 2$; the spectrum at state 3000 has no matching recorded gradient pair.

G. Controlled Adam Measurements and Spike Analysis

The Adam comparison reuses the complete objectives and frozen initial tensors specified above, with zero initial moments and a constant rate throughout each 1,024-update arm. All trainable parameters enter the diagnostics. Parameters, gradients and optimizer moments are FP32; geometric reductions are FP64, with TF32 disabled. No Hessian or Hessian-vector product is evaluated. MLP rates share one RTX 5070 Ti; ResNet uses qualified PRO 4000, PRO 6000-partition and RTX 4090 workers, and Transformer uses RTX 4090, PRO 4500 SE and RTX 4090, in increasing-rate order. Initial gradients are qualified across workers; subsequent finite-precision drift and hardware variation limit attribution exclusively to rate. Complete held-out splits are evaluated every 128 updates, without selecting rates on held-out loss; the main figures concern training loss.

A state s denotes parameters θ_s after s completed updates. The raw gradient g_{s+1} is evaluated there; m_{s+1} includes that gradient and is available before update $s + 1$. Figures 5 and 6 align these quantities to state s, avoiding an artificial one-update lead or lag. The overview omits the unapplied terminal gradient and uses complete 50-observation trailing arithmetic means of each pointwise diagnostic, with no padding. Norms are averaged before logarithmic display. The spike views contain every observation in their windows. A zero norm leaves cosine undefined; no zero denominator occurs in the reported pairs.

The loss-only episode analysis begins at state 64. An episode starts at a 10% rise above the minimum over the preceding 16 states, retains this fixed baseline through rebounds, and ends at the first loss strictly below it. Episodes still open at state 1,024 remain censored. There are 129 episodes, 45 reaching at least 50% above baseline, and two censored recoveries. The examples select the largest completed peak/baseline ratio at each highest-rate arm. Their plotting baselines are the losses ten states before the peaks, which can differ from the episode baselines.

For forecasting, eligible states $s \geq 64$ have loss less than 5% above the minimum over $s - 16, \ldots, s$. The target is any loss at least 10% above $\mathcal{L}(\theta_s)$ in the next eight updates. The baseline uses task/rate indicators, $s/1024$, log current loss, log-loss changes over one, four and eight states, and current log draw-up from the 16-state minimum. We compare it with the same features plus continuous projected gradient change. Fixed L2-logistic models use $C = 1$, training-only standardization and no class reweighting or hyperparameter search. Expanding training prefixes precede five test blocks beginning at states 384, 512, 640, 768 and 896. A 24-state gap ensures all training outcomes precede the earliest test loss-history input. Final states without complete future coverage are omitted. No random state split is used.

The 4,534 eligible forward-test states contain 495 positive targets. Average precision rises from 0.260 to 0.454 with projected change; Brier score falls from 0.11972 to 0.08685. Both improve in all five blocks. Crossing the horizons 4/8/16 with rise thresholds 5%/10%/25% retains an average-precision improvement in all nine settings. Adding gradient and momentum levels/growth together with projected change instead gives pooled average precision 0.395: more diagnostics do not uniformly improve the forecast. The fixed above-two rule has 704 false-positive states among 891 alarms. Episode-warning coverage and state-level alarm precision use different denominators and should not be equated.

The inspected Transformer example motivated this retrospective analysis. Later blocks are held out from fitting, but the analysis is exploratory; repeated states and episodes do not provide independent-run uncertainty. There is no intervention testing whether a diagnostic-guided rate change prevents a spike or improves subsequent loss.

TABLE III: Complete-prefix temporal accounting at $T = 3000$, with the quantities defined in Eq. (39). P_{3000} is path length; the last three columns are raw-gradient interactions. All nine conditions use one shared initialization per case. Values are rounded only for display, so displayed components may not sum exactly to the displayed total.

Model	η	P_{3000}	$I^{(1)}_{3000}$	$I^{(>1)}_{3000}$	I_{3000}
MLP	0.004	1.1976	1.270×10^{2}	5.634×10^{4}	5.647×10^{4}
MLP	0.02	64.3608	-1.217×10^{4}	3.063×10^{4}	1.846×10^{4}
MLP	0.04	185.5420	-1.662×10^{4}	2.225×10^{4}	5.630×10^{3}
ResNet-32	0.025	41.1785	-2.528×10^{3}	5.325×10^{4}	5.072×10^{4}
ResNet-32	0.05	146.3482	-7.380×10^{3}	3.603×10^{4}	2.865×10^{4}
ResNet-32	0.1	372.7896	-1.076×10^{4}	2.451×10^{4}	1.375×10^{4}
Transformer	0.05	15.6687	1.629×10^{2}	5.100×10^{4}	5.116×10^{4}
Transformer	0.1	111.8465	-7.767×10^{2}	2.464×10^{4}	2.387×10^{4}
Transformer	0.2	405.8404	-2.682×10^{3}	1.336×10^{4}	1.067×10^{4}

TABLE IV: Retrospective check of the local loss approximation for all nine conditions. The unthresholded count and median e_t use 2,999 transitions per condition. Each threshold cell gives opposite signs / eligible transitions; eligibility requires both change magnitudes to exceed $h_t(\tau)$. These descriptive temporal counts are not independent-sample predictive accuracies.

				Opposite signs / eligible transitions at threshold τ		
Model	η	No threshold	Median e_t	10^{-7}	10^{-6}	10^{-5}
MLP	0.004	0/2999	2.905×10^{-4}	0/2999	0/2999	0/2999
MLP	0.02	145/2999	6.146×10^{-4}	145/2998	145/2998	141/2989
MLP	0.04	281/2999	7.343×10^{-3}	281/2999	281/2998	280/2994
ResNet-32	0.025	595/2999	1.030×10^{-2}	595/2999	592/2991	570/2934
ResNet-32	0.05	694/2999	2.347×10^{-2}	694/2999	692/2997	682/2978
ResNet-32	0.1	814/2999	2.324×10^{-2}	813/2997	813/2997	809/2989
Transformer	0.05	0/2999	2.381×10^{-5}	0/2999	0/2999	0/2999
Transformer	0.1	542/2999	1.398×10^{-2}	541/2997	534/2977	436/2703
Transformer	0.2	1325/2999	6.323×10^{-2}	1324/2996	1319/2989	1239/2837

TABLE V: Final spectral context at parameter state 3000. Only the MLP measurement covers its complete training objective. ResNet and Transformer use fixed subsets of the stated size; residual-checked eigenpairs do not establish subset representativeness. All conditions use seed 0.

Model	η	Hessian objective coverage	$\eta\lambda_{\max,3000}$
MLP	0.004	Full 5,000 images	0.3168
MLP	0.02	Full 5,000 images	2.2157
MLP	0.04	Full 5,000 images	2.3259
ResNet-32	0.025	Fixed 5,000-image subset	2.8176
ResNet-32	0.05	Fixed 5,000-image subset	2.6599
ResNet-32	0.1	Fixed 5,000-image subset	3.2488
Transformer	0.05	Fixed 2,500-sequence subset	1.7229
Transformer	0.1	Fixed 2,500-sequence subset	2.2752
Transformer	0.2	Fixed 2,500-sequence subset	3.1202

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